Online Retail Data Mining

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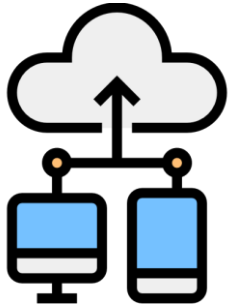
Almost all of our activity can be recorded as data.

Around 995 photos are uploaded on Instagram every second. (2020); Source: Omnicore Agency 2021.

Total Number of Tweets Sent per Day: 500 million. (2020); Source: Omnicore Agency 2021.

Roughly 1.01 billion credit card transactions occur every day around the world. (2018); Source: a research from The Nilson Report.



Even Intel predicted that there was 200 billion IoT devices by 2020.

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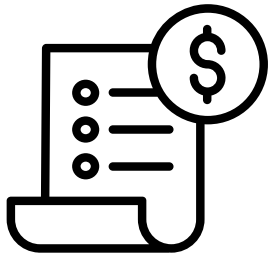
The same principle also applies to businesses: businesses collect data alongside doing their activities.



A common example: record of transactions (from receipts, etc.) between a business and its customers.

Accounting purposes

Legal purposes

Source = https://colemanandco-ni.co.uk/importance-recording-transactions/#:~:text=Recording%20transactions%20helps%20with%20many,you%20will%20gradually%20improve%20profit

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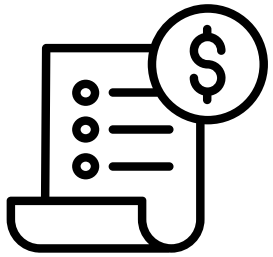
However, these data can be used for much more.



A common example: record of transactions (from receipts, etc.) between a business and its customers.

Customer behaviour segmentation

Fraud detection

Source = Chen, D. Sain, S.L., and Guo, K. (2012), Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining, Journal of Database Marketing and Customer Strategy Management, Vol. 19, No. 3, pp. 197-208. doi: [Web Link].

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Therefore, to present an use case of such data, I made a project about Customer behaviour segmentation with the following requirements:

Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01/12/2010 08:26	2,55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	01/12/2010 08:26	3,39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01/12/2010 08:26	2,75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01/12/2010 08:26	3,39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01/12/2010 08:26	3,39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01/12/2010 08:26	7,65	17850	United Kingdom
536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	01/12/2010 08:26	4,25	17850	United Kingdom
536366	22633	HAND WARMER UNION JACK	6	01/12/2010 08:28	1,85	17850	United Kingdom
536366	22632	HAND WARMER RED POLKA DOT	6	01/12/2010 08:28	1,85	17850	United Kingdom
536368	22960	JAM MAKING SET WITH JARS	6	01/12/2010 08:34	4,25	13047	United Kingdom
536368	22913	RED COAT RACK PARIS FASHION	3	01/12/2010 08:34	4,95	13047	United Kingdom
536368	22912	YELLOW COAT RACK PARIS FASHION	3	01/12/2010 08:34	4,95	13047	United Kingdom
536368	22914	BLUE COAT RACK PARIS FASHION	3	01/12/2010 08:34	4,95	13047	United Kingdom
536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01/12/2010 08:34	1,69	13047	United Kingdom
536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	01/12/2010 08:34	2,1	13047	United Kingdom
536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	01/12/2010 08:34	2,1	13047	United Kingdom
536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01/12/2010 08:34	3,75	13047	United Kingdom
536367	22310	IVORY KNITTED MUG COSY	6	01/12/2010 08:34	1,65	13047	United Kingdom
536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01/12/2010 08:34	4,25	13047	United Kingdom
536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	01/12/2010 08:34	4,95	13047	United Kingdom
536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	01/12/2010 08:34	9,95	13047	United Kingdom
536367	21754	HOME BUILDING BLOCK WORD	3	01/12/2010 08:34	5,95	13047	United Kingdom
536367	21755	LOVE BUILDING BLOCK WORD	3	01/12/2010 08:34	5,95	13047	United Kingdom
536367	21777	RECIPE BOX WITH METAL HEART	4	01/12/2010 08:34	7,95	13047	United Kingdom
536367	48187	DOORMAT NEW ENGLAND	4	01/12/2010 08:34	7,95	13047	United Kingdom
536369	21756	BATH BUILDING BLOCK WORD	3	01/12/2010 08:35	5,95	13047	United Kingdom
536370	22728	ALARM CLOCK BAKELIKE PINK	24	01/12/2010 08:45	3,75	12583	France
536370	22727	ALARM CLOCK BAKELIKE RED	24	01/12/2010 08:45	3,75	12583	France

Record of transactions (transaction data).

For this project, I use the open source transaction data made available by UC Irvine.

Open source python programming language libraries and IDE, such as pandas and Jupyter notebook.

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Project flow

The project is done through the template of CRISP-DM (Cross-industry standard process for data mining) framework: specifically the one mentioned in Page 75, Meta S. Brown, Data Mining For Dummies, Wiley Publishing Inc. (2014).

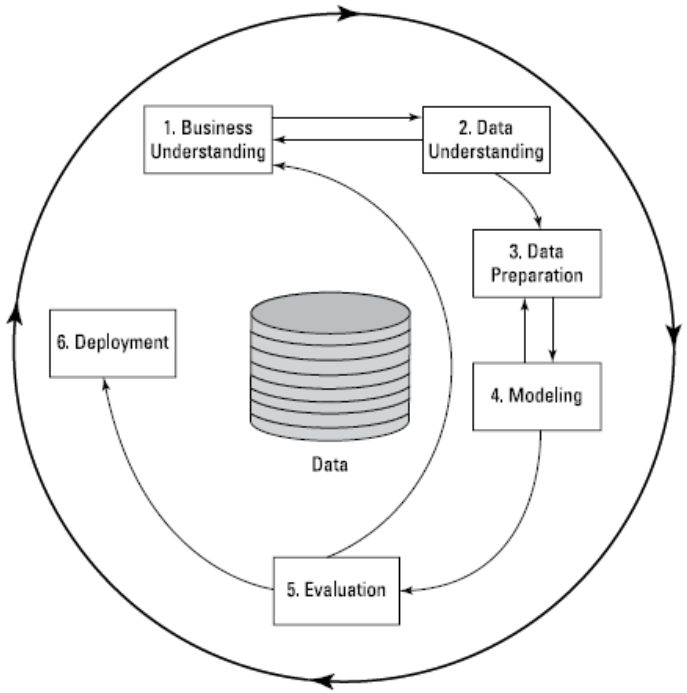
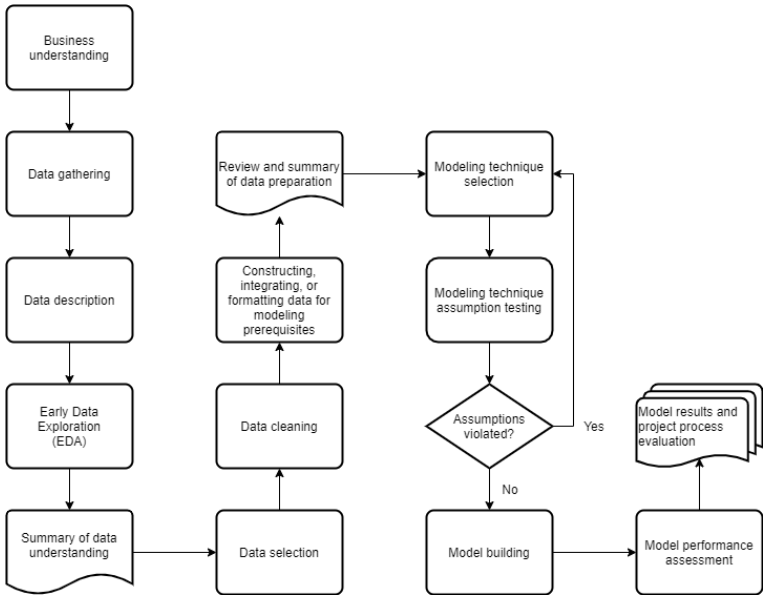


Figure 5-1: The CRISP-DM process model.



However, some modifications were made. On the left is a flowchart of the project's actual flow.

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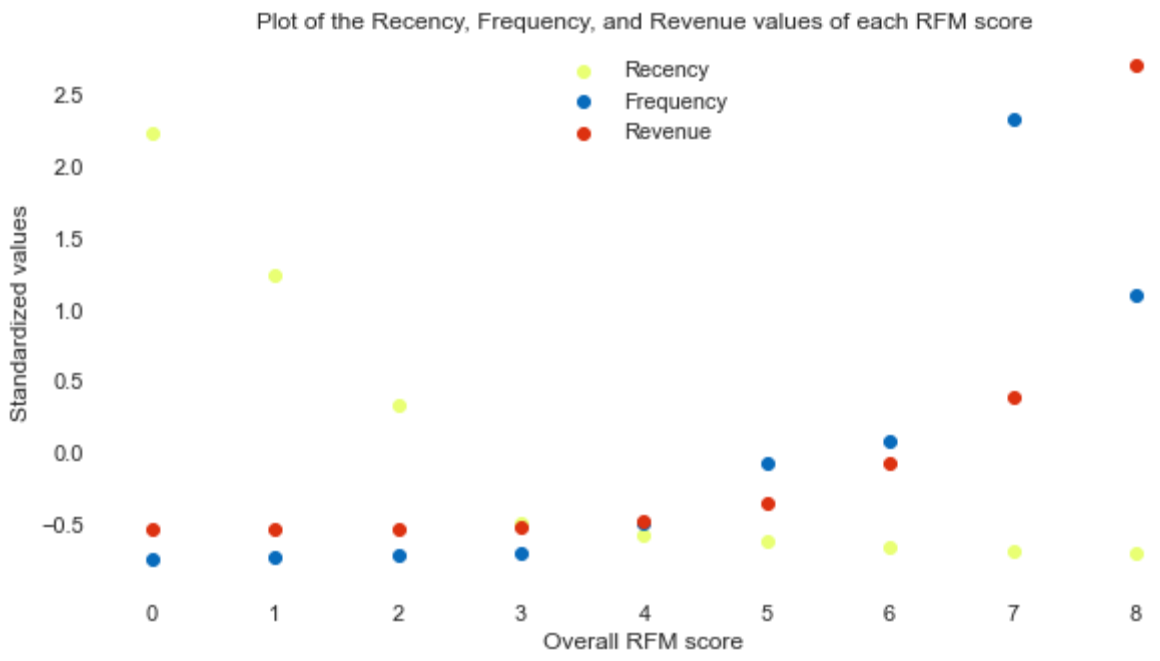
Project results

By calculating the RFM (Recency, Frequency, Monetary value) marketing metrics of each customers in the dataset and then implement the K-means clustering algorithm towards each metric's value, clusters of customers (each with their own unique behaviours) are identified.



The summary of the plot's analysis =

- ❑ Cluster 8: high spending customers that only regularly do transactions.
- ❑ Cluster 7: more-frequently-purchasing customers that are less willing to spend more money per transaction.
- ❑ Cluster 5-6: the medium spender and occasional customers.
- ❑ Cluster 0-4: once-in-a-while customers.

Afterword

To view the complete content of the study case, you can click the link below:

http://bit.ly/AndreasParasianOnlineRetailDataMining

Thank you and enjoy reading the case! Should there be any error from the assumptions or methods that I used, it would be very helpful and I would be very grateful to be notified about it. You can contact me through these channels:

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